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The Best Time to Redevelop a Mature Unconventional Oil Field to Maximise the Value

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Abstract

Towards the end of an unconventional field's life, declining production leads to reduced cash flows. However, there remains potential for value creation by drilling into isolated pockets or accessing remote areas of the reservoir from the original wells. Coiled tubing drilling (CTD) technology provides a promising solution for such oil and gas fields. It is more cost-effective and improves recovery compared to traditional drilling, although it carries a different risk and uncertainty profile. At this stage, the operator must make an important decision: continue production, abandon the field, or drill into remaining reserves using CTD. The timing of this decision is crucial, as it can have significant economic implications amid price uncertainty and operational challenges. Early infill drilling can accelerate production and potentially maximise value, especially when commodity prices are favourable. Conversely, delaying may be more advantageous if current prices are unfavourable. However, postponement might render it uneconomical or technically impossible.

This paper employs a real options valuation approach, demonstrated using a case study, to explore answers and sensitivities to this complex decision. The decision process is carried out using a flexible Markov chain, assuming that the optimal timing for re-entry or abandonment is affected by the mean-reverting nature of prices and the likelihood of technical success, which is assessed through a quantitative risk analysis. A recursive algorithm is used to identify the option that maximises the project's value compared to alternatives.

The results show that the unique features of CTD make it an ideal tool for redeveloping unconventional oil fields, especially during periods of rising prices. The key value drivers include accessing depleted reservoirs, improving recovery, and delivering a well in considerably less time and at a lower cost than traditional drilling rigs. However, the risk of not reaching the target due to wellbore collapse is an inherent threat associated with this technology. The example demonstrates that conducting infill drilling into remaining reserves before the field becomes heavily depleted can provide maximum value, particularly when capitalising on favourable commodity prices. Within the optimal timing, the field's value with an infill option surpasses that without one. The optimal redevelopment timing was identified using a decision analysis technique to enhance decision quality. The proposed model considers technical and economic

factors, delivering results swiftly. It also includes a sensitivity analysis to evaluate the impact of key variables.

Keywords: Mature oil fields, Infill drilling, Coiled tubing drilling, Markov decision process, Mean-reverting process

Introduction

As energy demand grows, oil and gas operators shift to more difficult resources in unconventional mature fields to meet increased needs, making redeveloping mature fields attractive because "It is easier to find oil or gas where you have already found it than to explore new fields" (Parshall, 2012). Mature fields are characterised by having already reached their production plateau, with many entering the decline phase and approaching the end of their economic viability. Currently, about 70% of the world's oil-producing fields fall into this category, and their share continues to grow steadily (O'Brien et al., 2016).

As oil and gas fields mature, substantial hydrocarbons are often left behind due to economic and technical challenges, particularly in complex geological structures, such as those with heavily fractured, depleted, or tight reservoirs. Oil companies predominantly use infill drilling to improve mature fields' recovery and accelerate production by increasing the flow rate and ultimate recovery (Alusta et al., 2011). This method utilises existing data on the distribution of unrecovered reserves and drilling expertise, enabling companies to minimise unexpected risks while efficiently and cost-effectively targeting new zones to recover bypassed oil. However, at some stage in a field's lifecycle, the cost of producing an additional barrel of oil exceeds the price the market is willing to pay. In typical situations, this leads to the abandonment of fields, leaving up to 70% of the oil unrecovered (Lake et al., 1992). Coiled tubing drilling (CTD) represents an advanced drilling method designed to reduce costs and enhance recovery, particularly at the end of field life. This technology can be used to access challenging remaining reserves in tight, depleted, heavily fractured and faulted reservoirs that conventional drilling can't reach. However, these methods are associated with inherent risks, such as failing to reach the target in the worst-case scenario, necessitating careful weighing of value against the risks in the planning stage before making the drilling decision (Alsharea, 2024).

The literature on infill drilling typically overlooks the value that can be generated from oil price fluctuations when determining the timing of the redevelopment option. The primary focus is usually on the geological uncertainty and considering oil price uncertainty as a source of risk, as mentioned by Babadagli (2019) and Syed, Negahban and Dahaghi (2021). It also engaged in comparative studies, such as those examining the value of infill drilling versus enhanced oil recovery (Alusta et al., 2011), infill drilling versus refracturing (Urban et al., 2016), underbalanced versus overbalanced drilling (Finley et al., 2006). These studies used the Net Present Value (NPV) to weigh the feasibility of the options, considering fixed economic parameters, including the oil price. The dynamics of the commodity price are essential to abandoning a field permanently, waiting, or pursuing reinvestment. Additionally, the timing of infill drilling is crucial for creating additional value from an unconventional mature field using CTD. Assessing these time-dependent factors requires a more sophisticated dynamic decision model, as traditional decision analysis methods are inadequate for measuring their impact. Moreover, oil price fluctuations are used to create value rather than being perceived as a risk.

This paper examines the dynamic value of infill drilling considering oil price uncertainty in mature oil fields using real options valuation. It also estimates the technical probability of success in reaching the target by CTD through a Monte Carlo simulation. This straightforward decision model, which consistently accounts for technical uncertainties and price variations, is implemented with an algorithm in Visual Basic for Applications (VBA). The proposed programme features an intuitive user interface, making decision-making easier. We incorporate insights from real-option valuation methods in the petroleum industry, such as Smith and Mccardle (1999) and Brandão, Dyer and Hahn (2005b), as well as research on abandonment options, such as Jafarizadeh and Bratvold (2012) and dynamic decision-making processes (Jafarizadeh and

Bratvold., 2019). The algorithm was implemented using VBA as a program with an easy user interface, following the approach used by Jafarizadeh and Bratvold (2019) to address the value of flexibility for hydraulic fracturing. However, infill drilling by CTD has a unique uncertainty profile compared to other methods, which was addressed using quantitative risk analysis. This straightforward decision model, which explicitly considers technical uncertainties and price dynamics, is crucial for making informed strategic decisions.

This study is organised as follows: Section 2 offers a concise review of real options valuation. Section 3 discusses the technical challenges in depleted oil fields and estimating the likelihood of accessing the remaining hydrocarbons. Section 4 details stochastic oil price models for project valuation. Section 5 introduces the Markov chain process for dynamic decision analysis. Section 6 demonstrates the implementation of the proposed real option valuation method on CTD applications in mature fields, with an illustrative example, and examines the results. Section 7 concludes the results.

The real options valuation

Real options thinking involves exploring strategic flexibilities that develop throughout a project, while real options valuation focuses on quantitatively assessing the specific value of that flexibility. Jafarizadeh & Bratvold (2009) provided further details about the contractual and operational flexibilities in the upstream sector, with examples that approach flexibilities using a dynamic way of thinking and the mechanics of real options valuation.

The real options literature depends on financial option pricing, assuming that a replicating portfolio of financial instruments can be created to imitate the benefits of the real options using the standard no-arbitrage principle. The real options concept was introduced in 1977 by Stewart Myers. However, the idea that considers the abandonment option of a mature field as an American-style option goes to Myers & Majd (1983).

Early real options literature in the oil industry assumed that the value of the field is linked to a single source of uncertainty, primarily tied to commodity price, usually modelled as the Geometric Brownian Motion (GBM). The same concept was used by Paddock et al., (1988) to address the best time to abandon an oil field permanently. The best abandonment time is believed to depend on estimated oil price and cumulative production, which are considered to follow a GBM. Dixit and Pindyck (1994) employed alternative solutions, incorporating dynamic programming and decision analysis to solve the real options problem.

Further advancement in real options methodology, Smith & Mccardle (1999) merged the option pricing approach with the decision analytic approach in oil project valuation. They differentiated between market uncertainties (such as oil prices) and private uncertainties (such as production and cost). They noted that market risks can be effectively hedged using traded financial instruments, while private risks cannot be perfectly hedged; therefore, they can be assigned a probability based on the insight of experts in the field. When both private and market risks have been addressed, the cash flow can be discounted using a risk-free discount rate. This approach is more consistent than the conventional risk-adjusted discount rate, which tends to underestimate the future value of marginal opportunities. However, some complex geological uncertainties that are affected by many factors can't be directly assigned a probability of occurrence. Moreover, relying on expert insight for quantifying technical uncertainty introduces the risk of bias (Knaut et al., 2012).

While the real options valuation approach has been widely adopted in the literature, modelling complexity has been noted as the main drawback, hindering its widespread adoption by the industry compared to the standard discounted cash flow (Jafarizadeh & Bratvold, 2009). With the advancement in computer capabilities, real options valuation has gained increased application in oil fields. The stochastic dynamic programming model has been used by Lund, (2000) to assess the decision for the redevelopment of an

offshore oil field, considering both market and private risks. They suggest limiting uncertain factors to only two or three realisations to maintain a manageable model size.

Copeland and Antikarov (2001) introduced a method to address real options problems by first calculating the expected NPV of the project without considering options. Then, a Monte Carlo simulation is used to model cash flow uncertainties and estimate key parameters. Finally, an event tree incorporating decision nodes and options is constructed to approximate the project's value over time, which is then solved using risk-neutral valuation. This approach is also known as the Marketed Asset Disclaimer (MAD). Instead of relying on the market-established discount rate for a project, the MAD approach incorporates subjective information to compute the option value. This reliance overlooks the potential to derive insights from capital markets. As a result, the MAD approach is most suitable for corporate investments with no clear financial market analogues.

Brandão et al. (2005) employed the MAD concept. However, they used a binomial decision tree model to represent the uncertainties in a project's value over time and a risk-neutral approach to determine the project value with options. This method is called BDH and is regarded as a milestone in real options and decision analysis, as it offers enhanced flexibility in problem modelling, including the ability to incorporate multiple uncertainties and concurrent options with intricate payoff structures. They achieved a straightforward solution by applying a binomial tree to approximate the GBM.

Smith (2005) extended the discussion on the BDH solution and proposed alternative approaches that use a fully risk-neutral valuation framework. Building on Smith's critique of the BDH method for solving real-options problems, Brandão et al., (2005) emphasised that no single technique is universally optimal for all real-options scenarios. They argued that the choice of modelling method should be based on the specific characteristics of the problem, the needs of the decision-maker, and the analyst's modelling preferences. However, a consistent model should yield the same results regardless of whether it is based on the binomial lattice or the binomial tree method. They also agreed that a fully risk-neutral approach is ideal when the required market information is available.

Jafarizadeh (2012) noted that merging a decision-tree and a risk-neutral approach offers a consistent framework for valuing flexible opportunities. However, decision trees and binomial lattices can be challenging when the investment decision depends on multiple uncertainties (such as oil prices and decommissioning costs) and involves several decision points. Therefore, they used the least-squares Montecarlo (LSM) method to assess the best time to abandon an oil field. Their work advances the real options literature by employing the Schwartz and Smith model and estimating its parameters through the implied approach, using options on futures.

Following the GBM, Pindyck (1999) confirmed the practicality of the mean reversion process in forecasting oil prices using 127 years of market data. This concept, which relies on the principles of supply and demand, has been widely accepted and adopted in the literature on real options valuation. Hahn & Dyer, (2008) developed a discrete version of the mean reversion process. This convenient model was the foundation for representing oil price uncertainty in a binomial lattice framework for real options valuation. For example, Jafarizadeh & Bratvold (2019) used this model to address the challenge of determining the optimal refracture time in shale reservoirs. The authors formulated the problem as a real option, where the value of any option depends on the mean behaviour of the oil price, represented in a binomial lattice within a risk-neutral approach. The decision is modelled as a Markov decision process, and the value of the best option is assessed using a recursive algorithm implemented with VBA. This paper relies on the mentioned concept in the decision-making process. However, it extends the model to account for three stages of a binomial lattice. It also quantifies the technical probability of success using a Monte Carlo simulation.

Technical uncertainty in depleted mature oil fields

Infill drilling is a standard option that can be undertaken at any stage during a field's lifespan. However, operators typically carry it out during the declining phase, when surface facilities have available capacity and the uncertainties surrounding the initial wells are clarified. The field operator can either accelerate cash flow during periods of rising prices (thus shortening the field's life) or wait until the original well's lifespan concludes before deciding on an infill campaign to prolong the field's economic viability. Permanent abandonment becomes an option when future redevelopment prospects appear unattractive. However, infill drilling becomes unviable due to economic factors, such as falling prices and the diminishing profitability of remaining reserves, or because geological conditions, like severe depletion, render the remaining reserves inaccessible. The value of an infill oil well is assessed by the discounted worth of its expected future production, which is closely linked to the outlook for future oil prices.

The remaining reserves in tight, heavily fractured, severely depleted, and faulted mature reservoirs are often considered non-commercial or complex resources, mainly due to the geological characteristics and the drilling problems (Johnson et al., 2008). When a mature field is heavily depleted, infill drilling with conventional overbalanced drilling (OBD) becomes difficult because of severe lost circulations, and the drill string is likely to become stuck and fail to reach the target. Alternatively, underbalanced drilling (UBD) offers a feasible means of reaching the required depth. Moreover, it minimises formation damage, which in turn improves incremental recovery.

The field applications of UBD worldwide show an increase in reserves from a minimum of 3% to a maximum of 50%, compared to the conventional OBD wells drilled in the same fields (Al-Sharea, 2024). The same references mentioned that CTD enables access to these remaining reserves at an average cost saving of 35% compared to conventional rigs, as it is specifically designed to manage underbalanced drilling conditions effectively. However, wellbore collapse is a common risk that can lead to pipe sticking and fishing, representing a worst-case scenario of the underbalanced conditions. This problem occurs when the drilling fluid's Bottom Hole Pressure (BHP), which is required to prevent wellbore instability, becomes less than the rock collapse pressure. Predicting collapse, BHP, and fracture pressure are subject to inherent uncertainties. Aadnøy (2011) noted that the uncertainty of wellbore instability can be consistently assessed when an informed range of input parameters is defined. Moos et al.(2003) and Udegbumam, Aadnøy and Fjelde (2014) have addressed this problem using quantitative risk analysis (QRA).

This section uses the informed range of input parameters mentioned by Aadnøy (2011) to quantify the probability of success or failure in reaching the target. It uses an elastic model to predict collapse pressure (P_c) and the informed range of uncertainty of BHP. The informed range of fracture pressure (P_f) is used here to ensure physically consistent results. If the results were inconsistent, the input range of magnitude should be redefined. Fig. 1 indicates that with an informed range of input parameters, there is a 20.5% probability that BHP will fall below the 5th percentile of collapse pressure (P_c), measured at a specific gravity of 1.546. Although geological risk cannot be eliminated, quantifying its probability guides strategic decisions, such as whether to proceed with infill drilling.

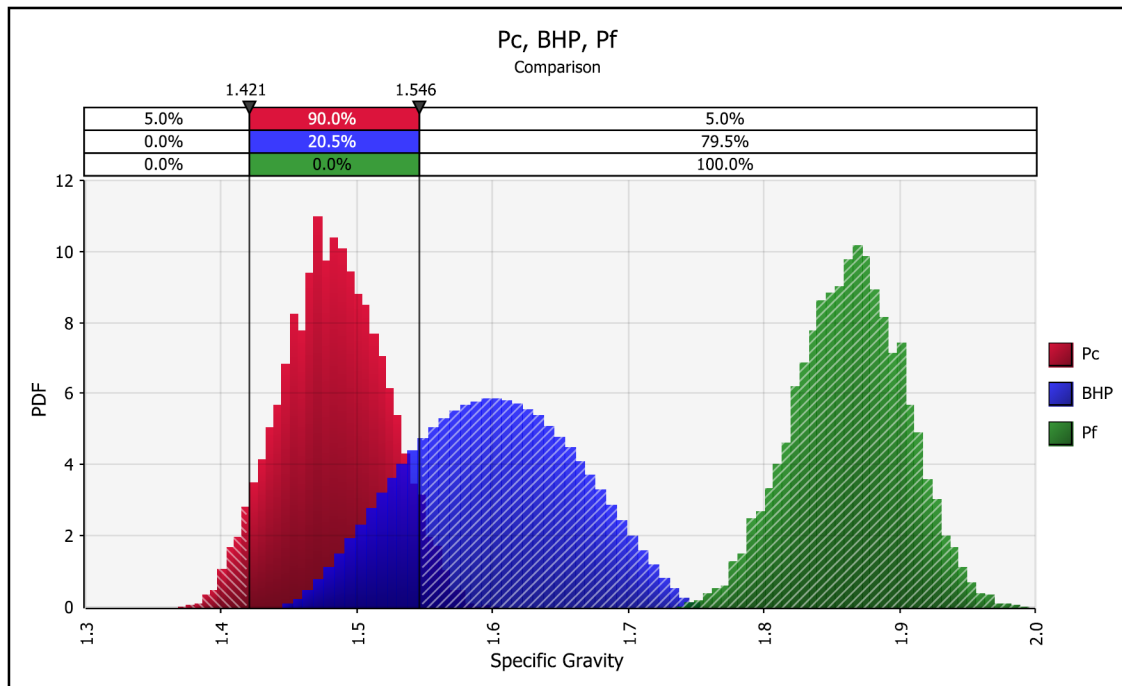


Figure 1—The probabilistic approach to determining the probability of wellbore collapse is expressed as the probability of bottom hole pressure (BHP) crossing the wellbore collapse pressure (Pc) or (shear failure pressure). BHP is the safe drilling window between collapse pressure (Pc) and fracture pressure (Pf). A Monte Carlo simulation is performed over 10,000 times by commercial software.

Oil Price model for project valuation

The economic worth of an oil well depends on future price expectations. As the commodity price increases, the drilling option becomes more valuable; conversely, it becomes uneconomical when prices fall. Maintaining consistent confidence in the price outlook is crucial for making informed decisions. Oil price stochastic models are mathematical tools that forecast how prices may evolve over time. Early models of oil prices assumed they could be represented by a Geometric Brownian Motion (GBM) process. This analogy draws on the behaviour of stock prices in capital markets. The widely known Black-Scholes equation underpins the GBM. GBM is often used in real options valuation for its simplicity and the need for a few parameters when modelling oil prices. Eq. 1 mathematically represents the stochastic process of GBM.

$$dP = \alpha P dt + \sigma P dz \quad \text{Eq. 1}$$

Where: dP is the change in price over time. α is the drift. P is the current oil price. dt is the change in time. σ is the volatility. $dz = \varepsilon \sqrt{dt}$, ε is the Wiener process, which is defined by a standard normal distribution, expressed as $N(0,1)$, with a mean of zero and a standard deviation of one.

To apply GBM, two key parameters must be estimated: the drift (α) and the volatility (σ). For instance, Dias (2004) mentioned that volatility typically ranges between 15% and 25% annually. He also noted that other studies have used an annual volatility value of approximately 30%. An alternative method for estimating volatility involves using Monte Carlo simulations to model the stochastic processes that drive key market variables. Geometric Brownian Motion (GBM), or random walk, is highly effective for short-term valuation due to its ability to capture volatility. However, it overlooks other essential time-series components such as trends, seasonality, and structural patterns, making it less suitable for long-term project valuation. Furthermore, unlike mean reversion models, GBM exhibits divergence over the project's lifespan, which leads to high upper and lower levels of confidence in extended time horizons (Jafarizadeh, 2022). This behaviour is as shown in Figure 2.

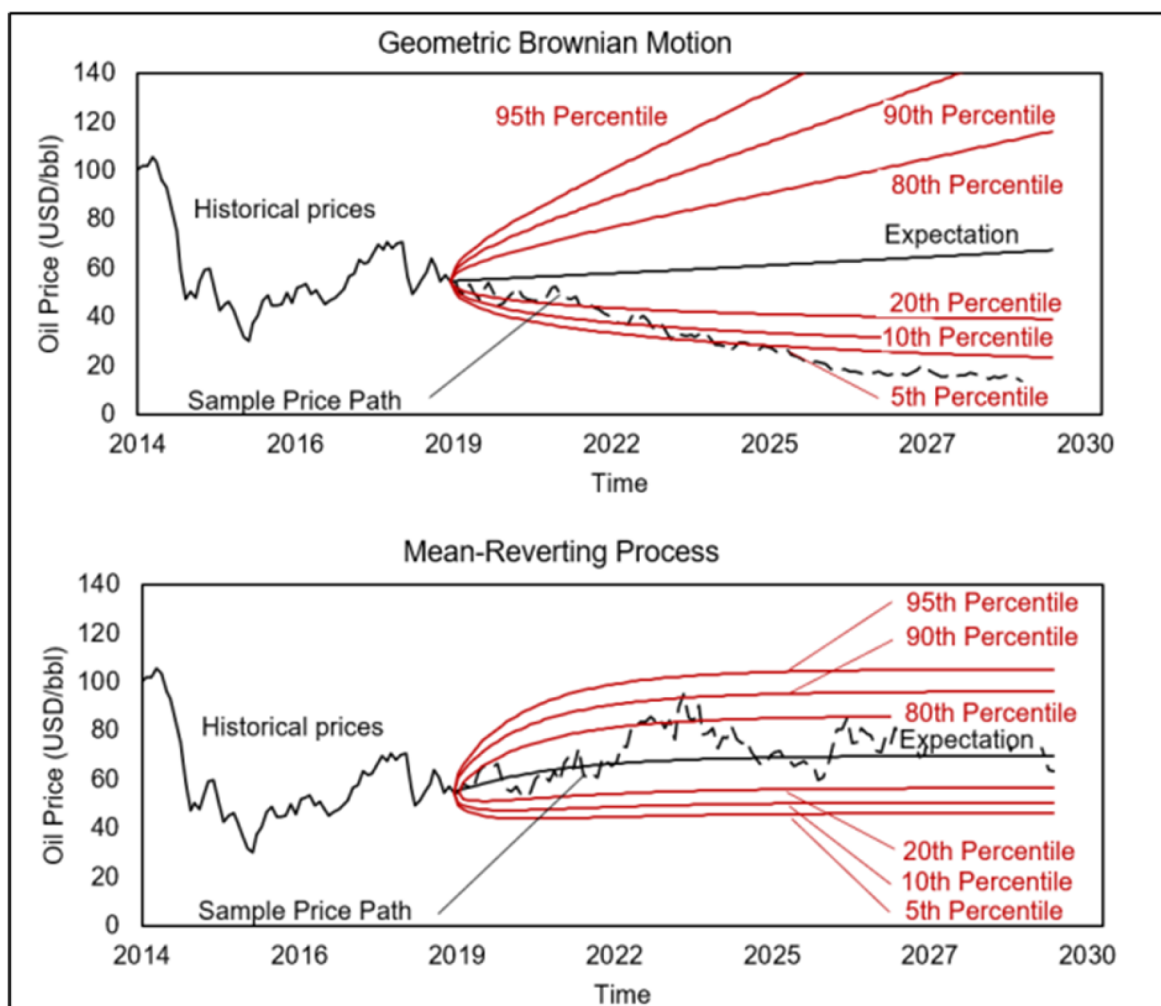


Figure 2—mean reversion process versus the GBM model (Jafarizadeh, 2022).

While various sophisticated models exist for forecasting commodity prices, the mean-reverting price process remains particularly attractive both in theory and in practice (Jafarizadeh and Bratvold, 2019). In equilibrium, rising prices encourage high-cost oil producers to enter the market, expanding supply and gradually driving prices back toward the mean. Conversely, when prices fall, these higher-cost producers exit, tightening supply and pushing prices upward. This cyclical interaction between price, supply, and demand underpins the mean-reverting behaviour commonly observed in oil markets. Examples of using the mean reversion model in the oil industry include those by Smith & Mccardle (1999), Dias (2004), and Jafarizadeh and Bratvold (2019). Within the GBM framework, any change in oil prices is viewed as a lasting shift in the long-term trend. By contrast, the pure mean-reversion model treats price movements as short-term deviations around a stable long-run equilibrium. A more balanced approach combines these perspectives, capturing the stochastic nature of prices through GBM while reflecting long-term stability through the mean-reversion model (MRM). Schwartz & Smith, (2000) presented a well-recognised two-factor model to simulate this process. Their model incorporates mean reversion in the short term while accounting for long-term uncertainty or permanent effects after prices revert to their average value. Long-term behaviour is modelled using Brownian motion, considering new explorations, technological advancements, inflation, and political and regulatory impacts. Conversely, short-term variations from balance prices are modelled as a mean-reverting change that follows the Ornstein-Uhlenbeck. This model can be easily converted into a single-factor mean-reversion model.

Estimating model parameters is essential and challenging. Although [Schwartz & Smith \(2000\)](#) used the Kalman filter, they acknowledged that market-implied estimates can be used to determine model parameters. Instead of estimating parameters from historical price data, the implied approach calibrates model parameters with futures so that model prices of oil derivatives match observed market prices. Unlike the Kalman filter technique, the implied approach to parameter estimation is straightforward and representative, providing reasonable estimates for valuation assessments ([Jafarizadeh & Bratvold, 2012](#)). They also noted that futures and options on futures are risk-free instruments intended to hedge against price fluctuations. Thus, the risk-neutral model with implied parameters similarly provides a risk-neutral representation of prices.

In the Schwartz and Smith model, the mean-reverting component is represented by a constant long-term average and a short-term stochastic deviation. These deviations oscillate randomly around zero, where positive values reflect spot prices above the average and negative values indicate prices below it. Formally, the spot price S_t at time t is expressed as $\ln(S_t) = \chi_t + \xi$, $\xi_t(x_i)$ denoting the time-invariant long-run equilibrium level, while the short-term deviations χ_t evolve according to an Ornstein–Uhlenbeck process, as shown in [Eq. 2](#).

$$d\chi_t = -\kappa\chi_t dt + \sigma dz \quad \text{Eq. 2}$$

In this context, κ denotes the mean-reversion coefficient, σ represents the volatility of the short-term factor, and dz corresponds to the increment of a standard Brownian motion. A larger mean-reversion coefficient indicates that short-term deviations return to equilibrium more quickly, whereas higher volatility reflects greater fluctuations around the long-term average price. Spot prices are not directly useful for project valuations; what is needed instead are price forecasts, which can be derived from expectations of future market conditions ([Jafarizadeh and Bratvold, 2012](#)). The value of a forward contract ($F_{t,T}$), which specifies delivery of a barrel of oil at time (T) agreed upon at time (t), is derived from the spot price and is therefore affected by the variable χ_t , as expressed in [Eq. 3](#).

$$F_{t,T} = e^{-k(T-t)}\chi_t + \xi + (1 - e^{-2k(T-t)})\frac{\sigma^2}{4k} \quad \text{Eq. 3}$$

[Hahn & Dyer, \(2008\)](#) developed a binomial-lattice formulation for representing a continuous mean-reverting process. Over a time step (Δt), the price variable χ may increase to (χ_t^+) with an upward probability (p_u) or decrease to (χ_t^-) with a downward probability (p_d), these probabilities were restricted to lie between zero and one, as defined in [Equations \(4 – 7\)](#).

$$\chi_t^+ = \chi_t + \sqrt{\Delta t}\sigma \quad \text{Eq. 4}$$

$$\chi_t^- = \chi_t - \sqrt{\Delta t}\sigma \quad \text{Eq. 5}$$

$$p_u = \max\left(0, \min\left(1, \frac{1}{2} - \sqrt{\Delta t}(k\chi_t + 1/2\sigma^2)/2\sigma\right)\right) \quad \text{Eq. 6}$$

$$p_d = 1 - p_u \quad \text{Eq. 7}$$

In this paper, the oil price is modelled as a binomial lattice. The forward curve starts from each node, representing the price forecast for ten years. We fitted our model using futures contracts to estimate the price parameters, following the approach mentioned by [Jafarizadeh and Bratvold \(2012\)](#). Spot price movements are depicted as solid lines across three-time steps, each spanning one year. Dashed lines illustrate price forecasts extending to year 10, originating from each node. The green solid line represents the long-term expected oil price estimated using the mean reversion process starting from the initial point, as shown in [Fig. 3](#).

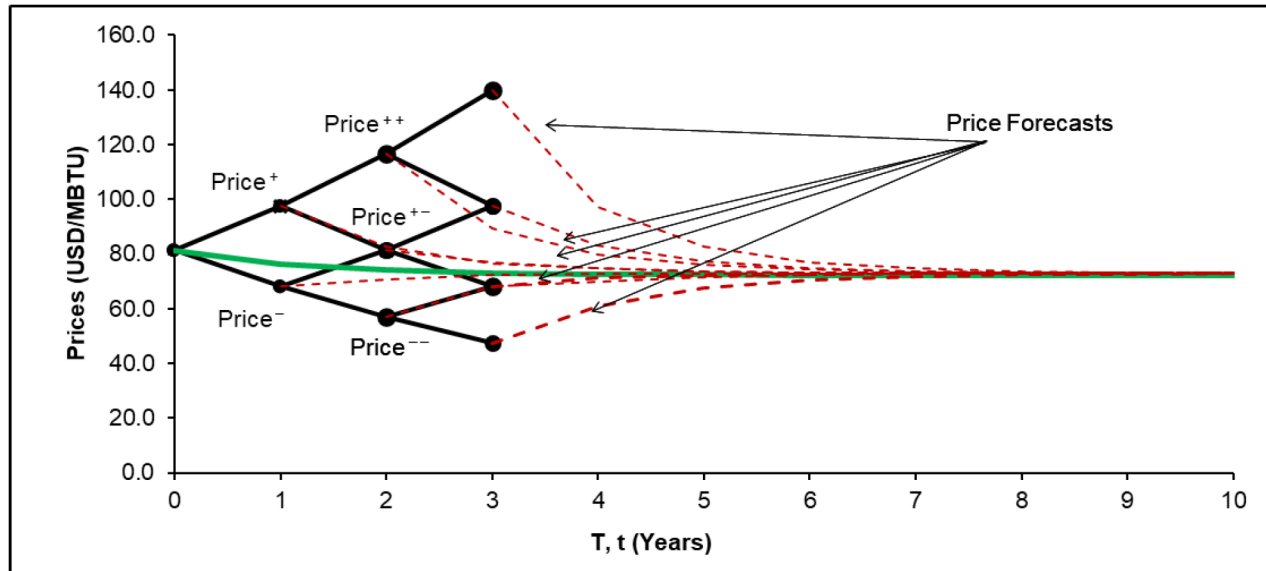


Figure 3—The binomial-lattice representation of oil price for the short term and continuous mean-reverting process for the long term.

In each time step (year), the probabilities depend on (χ_t) , which reflects the deviation from the mean, as mentioned in equations (4) and (5). Eq. 6 further restricts these probabilities to lie between zero and one. The decision-making process starts once production begins to decline and the infill drilling option becomes available, continuing until the economic limit is reached. During this timeframe, operators evaluate the feasibility of their choices on an annual basis, taking into account oil price uncertainty, remaining reserves, and the likelihood of successfully accessing them through drilling.

At the starting point of the binomial lattice, t equals tau (τ), where (χ_τ) is expressed as a function of time (τ) with an initial value of χ_0 . Since the mean level (ζ) is constant and independent of time, future price movements are driven solely by short-term deviations. Thus, (χ_τ) can be used to represent future prices, where $\chi_\tau - \chi_0 = \ln F_{0,\tau} - \ln F_{0,0}$, $F_{0,\tau}$ is obtained from Eq. 3. Using Eq. 9, which is derived from Eq. 8, the value of (χ_τ) at the outset of the binomial lattice can be determined. Subsequently, Equations (4–7) are applied to build the binomial lattice.

$$\chi_\tau = e^{-k\tau}\chi_0 + \zeta + (1 - e^{-k\tau})\frac{\sigma^2}{4k} - \chi_0 - \zeta + \chi_0 \quad \text{Eq. 9}$$

$$\chi_\tau = e^{-k\tau}\chi_0 + (1 - e^{-k\tau})\frac{\sigma^2}{4k} \quad \text{Eq. 10}$$

Decision Structure: Markov chain

The decision tree used to structure embedded real options. The initial decision involves permanent abandonment, re-entry drilling, or continuation. Managers analyse the remaining reserve value based on estimated commodity prices at specific stages and make their decisions. If they decide to continue, they reassess prices in the next period and face the same choices. The open-ended lines indicate continuation, meaning the oil price will be updated the following year, and the same options will be faced. If the economic outlook for the drilling option is favourable, the algorithm advises conversion; otherwise, it recommends proceeding to the next decision point for further evaluation. This process continues until the drilling window closes, as shown in Fig. 4.

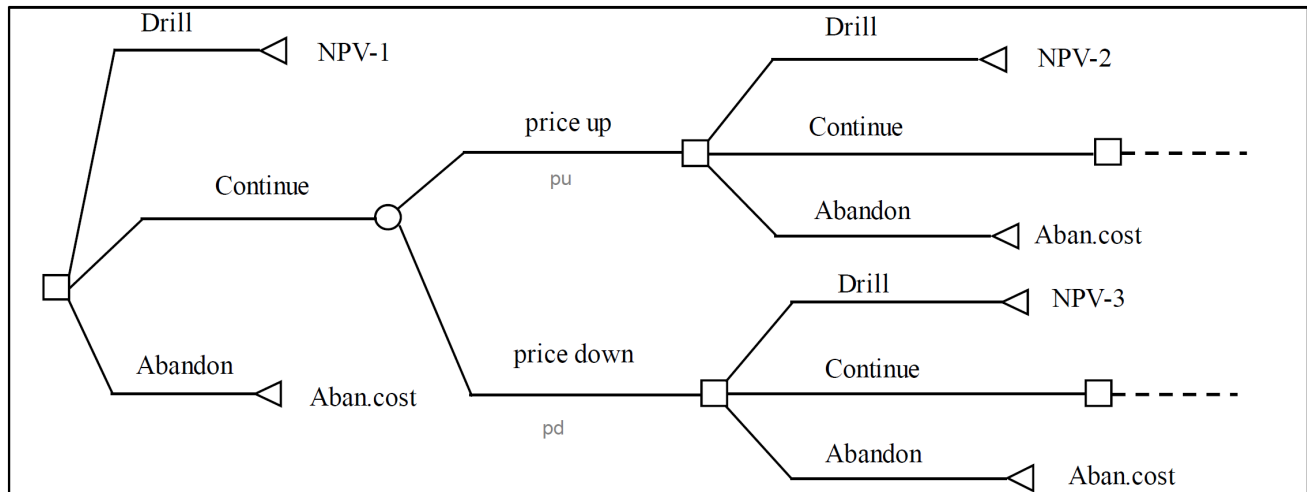


Figure 4—The dynamic decision approach, including the decisions and their updated probability of occurrence and expected outcomes.

Comparing the value of all alternatives is essential to determining the optimal course of action at each decision point. Therefore, the decision depends not only on the immediate value of an action but also on the potential impact of subsequent decisions and their outcomes. The recursive algorithm works backwards to identify the NPV of the best option.

Re-entry opportunities in mature fields exist within a limited timeframe, in our case, which is defined by the opening time (τ) and closing time (T). As noted earlier, drilling re-entry too early or too late may be impractical due to geological constraints or economic considerations. Extending field life through re-entry drilling not only generates additional production but also postpones abandonment costs. Conversely, if prospects appear unfavourable, immediate abandonment remains a viable option. The economic life of wells is evaluated against an economic threshold, which depends on projected oil prices. In this study, futures contracts are treated as risk-neutral forecasts of future price trends and are used to calibrate the parameters of the mean-reversion process. It is further assumed that managers rely on forward curves to evaluate the expected discounted payoffs of their investment decisions.

In a mean-reverting process, the forward curve is driven by the parameter (χ_t), which enables the definition of a deterministic payoff function $g(\chi_t)$ at each decision point. Since (χ_t) follows a Markov process (meaning its current value is independent of its past) the option value at any time (V_t) also depends solely on (χ_t). This is expressed as $v_t(\chi_t)$, for $V_t = v_t(\chi_t)$, $t = \tau, \tau + 1, \tau + 2, \dots$. As the opportunity window nears its closing time ($t = T$), and with production declining, managers must choose the best option, whether to continue operations, abandon the field, or pursue re-entry drilling, thereby obtaining $g(\chi_T)$. For periods within the window ($\tau < t < T$), managers compare the immediate exercise payoff ($g(\chi_t)$) against the expected value of deferring the decision to the next year. Starting from T , the valuation process is solved recursively by the algorithm, as outlined in Eq. 10 and Eq. 11.

$$V_T(\chi_T) = \max(g(\chi_T), NPV^{\chi_T} \text{Continue}) \quad \text{Eq. 11}$$

$$V_t(\chi_t) = \max\left(g(\chi_t), \frac{1}{(1+r)} (Pu v_{t+1}(\chi_t^u) + pd v_{t+1}(\chi_t^d)) + CF_t\right) \quad \text{Eq. 12}$$

For $\tau < t < T$, r denotes the discount rate, and $g(\chi_t)$ represents the NPV of the available options—either abandonment or re-entry—at time (t). The parameters τ and (T) correspond to the opening and closing times of the opportunity window, respectively.

The value of the option that has the maximum NPV is determined using Eq. 12.

$$g(\chi_t) = \max(NPV^{\chi_t}_{Abandon}, NPV^{\chi_t}_{Reentry}) \quad \text{Eq. 13}$$

Let (NPV^{χ_t}) represent the net present value at time (t) of the total cash flows associated with a particular course of action, conditional on (χ_t) . If the manager chooses to continue operations, (CF_t) corresponds to the cash flow generated by the production profile of the existing wells. If the decision were to drill, the production of the new well would be added to the cash flow calculation. The recursive algorithm in Eq. 10 determine the best time of re-entering compared to abandonment, and Eq. 10 determine the value of abandoning the oil field compared to continuing at $(t = \tau)$.

The total project value (V_0) , consists of the option value $(v_t(x_t))$ at the start of the opportunity window plus the discounted cash flows accrued before the option period $(0 \leq t < \tau)$, as expressed in Eq. 13.

$$V_0 = \sum_{t=0}^{\tau} \frac{CF_t}{(1+r)^t} + \frac{v_t(\chi_t)}{(1+r)^{\tau}} \quad \text{Eq. 16}$$

The system evolves according to prices following a mean-reverting process, with the stopping rule offering two options: either abandon the field immediately or extract the remaining reserves now while deferring abandonment costs. In this context, estimating the associated costs, benefits, and probability of reaching the required depth is complex. Therefore, we rely on the relevant literature to estimate the cost savings and the value of reducing formation damage achieved by the underbalanced drilling method. For simplicity, in the resolving algorithm, we assume a technical success probability of 1.

Illustrative Example

This example illustrates the valuation process for a re-entry well (open hole) believed to be drilled in an onshore oilfield in the United States of America. The initial oil well is in the decline phase and is expected to produce for six years, with the option to re-enter or abandon it during the second, third, and fourth years of decline. In our assessment, we consider specific technical and economic assumptions as detailed below.

Technical parameters

The initial well starts with a production of 24,600 barrels per year, declining at a rate of 40% annually. Its economic limit is projected to be reached in the sixth year. The re-entry well targets an isolated reservoir pocket and is assumed to follow the same 40% annual decline rate.

The drilling cost, which is the only capital expenditure (CAPEX) for the re-entry well, amounts to \$1 million, reflecting the typical cost of open-hole drilling in a U.S. onshore field. This includes a 35% cost reduction achieved through Coiled Tubing Drilling (CTD) compared to traditional methods.

Operational expenditure (OPEX) is estimated at \$22 per barrel, based on the average for U.S. onshore fields as reported by the 2016 U.S. Energy Information Administration (EIA).

Abandonment costs are set at \$160,000 per well, corresponding to the p90 estimate for onshore U.S. wells (Raimi et al., 2021). Deferring abandonment for five years yields a present value of approximately \$25,483, assuming a risk-free compound interest rate of 3% annually.

Economic parameters

Oil prices are estimated using the Schwartz and Smith mean-reverting model to capture long-term trends. Model parameters are calibrated by fitting to futures contracts data. A risk-free discount rate is applied, reflecting the risk-neutral valuation (Smith and Mccardle, 1999). The valuation relies on a recursive algorithm, updated from the methodology presented by Jafarizadeh and Bratvold (2019). In this case of an unconventional reservoir, production declines sharply following the plateau phase, requiring managers to evaluate three potential actions: abandon the field, maintain current operations, or re-access the reservoir using CTD. The key considerations are the probability of successfully reaching the remaining reserves

and the uncertainty of oil prices. The central question is: when should action be taken, and which option maximises the project's value?

Traditional decision tree

The conventional decision tree approach offers insight into the best option. However, it doesn't capture the flexibility that evolves with dynamic oil prices. In other words, it doesn't tell when the best time is to act and the value of deferring the abandonment cost, as shown in the in Fig. 5.

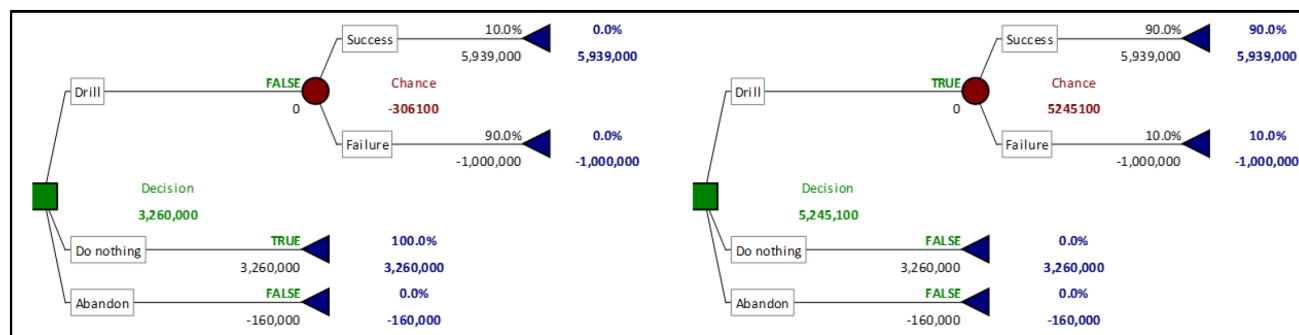


Figure 5—The traditional decision tree incorporates options, technical probabilities, and outcomes.

The decision tree on the left shows that the "do nothing" option has an expected value of 3,260,000 USD, compared to the drilling option, which has a negative expected value of (-306,100 USD). Doing nothing (continuing with the original wells) is the best choice when the probability of reaching the target through re-entry drilling is only 10%. If the drilling option fails to reach the target, it results in a loss of \$1,000,000, which is the cost of drilling. The decision tree on the right indicates that the drilling option has an expected value of \$5,245,100 when the probability of success increases to 90%. Doing nothing still has an expected value of 3,260,000 USD. Drilling is the better choice compared to doing nothing. The abandonment cost will also be deferred when the drilling option is chosen. The value of delaying the abandonment cost (\$160,000) for five years is \$25,483, assuming a risk-free annual interest rate of 3%. This amount is added to the field's value when the drilling option is selected, making the total value \$5,270,583. While this approach provides insight into the best option, it does not capture the value of flexibility that managers utilise. Moreover, the sensitivity analysis of the parameters needs to be assessed.

Real options valuation

The program interface of the proposed real option framework, which considers this flexibility in decision-making, is shown in Fig. 6.

Fig. 6 illustrates the program interface, which involves annual production, cost (capex and opex), the parameters of the stochastic price model (the mean reversion coefficient (κ), the standard deviation (σ), the long-term factor (ξ) and short-term deviations (χ_i)) and option parameters (discount rate, re-entry open and close time, and abandonment cost). The output includes "Well NPV + option", representing the project value with the re-entry option, whereas "Well NPV no option" reflects the value of continuing with the original wells. For simplicity, the technical probability of success is assumed to be 1 (100%). As mentioned in Section 3, the probability can be estimated stochastically using geomechanical models.

Price Parameters		Project Specifications										
χ_0	0.29	Time	0	1	2	3	4	5	6	7	8	9
ξ	4.27	Price	95.6	81.8	76.3	74.0	73.0	72.6	72.4	72.3	72.3	72.2
σ	18%	Production	24.6	14.8	8.9	5.3	3.2	1.9	1.1			
k	0.83	Cost	615.9	369.6	221.7	133.0	79.8	47.9	28.7			
Option Parameters		Re-entry 1			14.8	8.9	5.3	3.2	1.9	1.2		
Discount rate	5%	Costs 1			1370.0	222.0	133.2	79.9	48.0	28.8		
Re-entry window open	2	Re-entry 2			14.8	8.9	5.3	3.2	1.9	1.2		
Re-entry window close	4	Cost 2			1370.0	222.0	133.2	79.9	48.0	28.8		
Time step	1	Re-entry 3				14.8	8.9	5.3	3.2	1.9	1.2	
Abandonment Cost	160	Cost 3				1370.0	222.0	133.2	79.9	48.0	28.8	
Parameters of Sensitivity Analysis		Total	59.9									
	Min	Max										
χ_0	-0.3	0.3	Re-entry 1	74.7	25%							
ξ	3.5	4.5	Re-entry 2	83.6	40%							
σ	10%	50%	Re-entry 3	88.9	48%							
k	0.1	2.8										
Discount rate	0%	15%										
Sensitivity of Successful Re-entry												
	Min	Max										
P_{success}	0	1										
Re-entry valuation												
Results												
Well NPV + Option		5939.14										
Well NPV no Options		3259.55										

Figure 6—The program interface of the valuation model for the re-entry option; all values are multiplied by 1000.

Following the process outlined in Equations (4 – 7), constructing the binomial lattice shows that (χ_t) starting at 0.13, it can increase to 0.31 or fall to -0.05 in the first year. In the second year, the value of 0.31 can reach 0.49 or fall to 0.13, and the value of -0.05 can increase to 0.13 or decrease to -0.23 . In the third year, there are four probabilities for the (χ_t) with a maximum of 0.67 and a minimum of -0.41 . Accordingly, the oil price (\$/bbl) that starts at 81.5 can go up to 97.5 or down to 68 in the first year. In the second year, it can increase to 116.7 or fall to 56.8. The highest and lowest levels in the third year are 139.8 and 47.5, respectively, representing the best and worst scenarios, as shown in Fig. 7.

Year	0	1	2	3	Year	0	1	2	3	year	0	1	2	3
χ_t	0.13	0.31	0.49	0.67	$\chi_t + \xi_t$	4.4	4.58	4.76	4.94	Prices	81.5	97.5	116.7	139.8
		-0.05	0.13	0.31			4.22	4.4	4.58			68.0	81.5	97.5
			-0.23	-0.05				4.04	4.22				56.8	68.0
				-0.41					3.86					47.5

Figure 7—Estimating oil price using a binomial lattice.

What is the best time to drill, and what option maximises the project value?

The decision process mimics the scenarios analysed by a decision analyst over three years. It begins at the decision point and illustrates the likelihood of rising or falling oil prices. The binomial lattice is used to determine the outcome of each option. The future value of each choice is estimated using the forward curves originating from each point, enabling decision-makers to select the best timing to maximise project value. Using this methodology, the value of the re-entry option is determined as follows. At $t = 1$, (χ_t) begins from 0.13. The evolution of (χ_t) is then modelled through a binomial lattice using Eq. 4 to Eq. 7. By $t = 4$, after three time intervals, (χ_t) could vary between a minimum of -0.41 and a maximum of 0.67. Based on these values, forward curves are constructed for each (χ_t) to estimate the value of each possible decision. Applying the recursive algorithm described in Eq. 10 to Eq. 12, the option value at the initial decision point is calculated to be \$3,570,000, as illustrated in Fig. 8.

NPV (The value of the best)				Chi-lattice				NPV (Re-entry option)				NPV (Continue)			
t1	t2	t3	t4	t1	t2	t3	t4	t1	t2	t3	t4	t1	t2	t3	t4
3570.40	4003.35	1718.23	311.74	0.13	0.31	0.49	0.67	0.00	4003.35	1718.23	311.74	0.00	914.77	511.90	248.53
0.00	3705.70	1529.11	191.50	0.00	-0.05	0.13	0.31	0.00	3705.70	1529.11	191.50	0.00	774.27	422.01	191.41
0.00	0.00	1360.32	141.53	0.00	0.00	-0.23	-0.05	0.00	0.00	1360.32	84.34	0.00	0.00	343.35	141.53
0	0	0.00	97.90	0.00	0.00	0.00	-0.41	0.00	0.00	0.00	-11.44	0.00	0.00	0.00	97.90

Figure 8—The result of the binomial lattice for chi lattice, re-entry option, continuation, and the best option.

The figure also demonstrates that the re-entry option has a higher value than the continuation, and the best time to drill is in year 2 because this option maximises the project's value. In year 2, the maximum value of the re-entry option is determined to be 4003.35 USD, and the minimum is 3707.7 USD. Accordingly, this is the most advantageous value compared to continuing and drilling in another year (year 3 or 4). In the same year, the maximum value of continuing is 914.77 USD, and the minimum is 774.27 USD.

Sensitivity analysis

The two-dimensional sensitivity analysis demonstrates how key factors affect the project value. The results show that drilling a re-entry well using UBCTD is economically viable only when the success probability exceeds 50%; below this level, continuing with the original production strategy is more beneficial. The breakeven point indicates where the re-entry option yields a higher NPV than maintaining the declining output of the original wells. The discount rate has a limited impact on the overall project value due to the rapid decline in reservoir production, which coincides with the sharp fall in oil prices in this scenario. Since most oil is produced during the early stages, later cash flows are less affected by discounting. The project value is also influenced by oil price uncertainty, represented by volatility and the mean reversion coefficient. Higher volatility boosts the project value, while a lower mean reversion coefficient also increases value, as shown in Fig. 9.

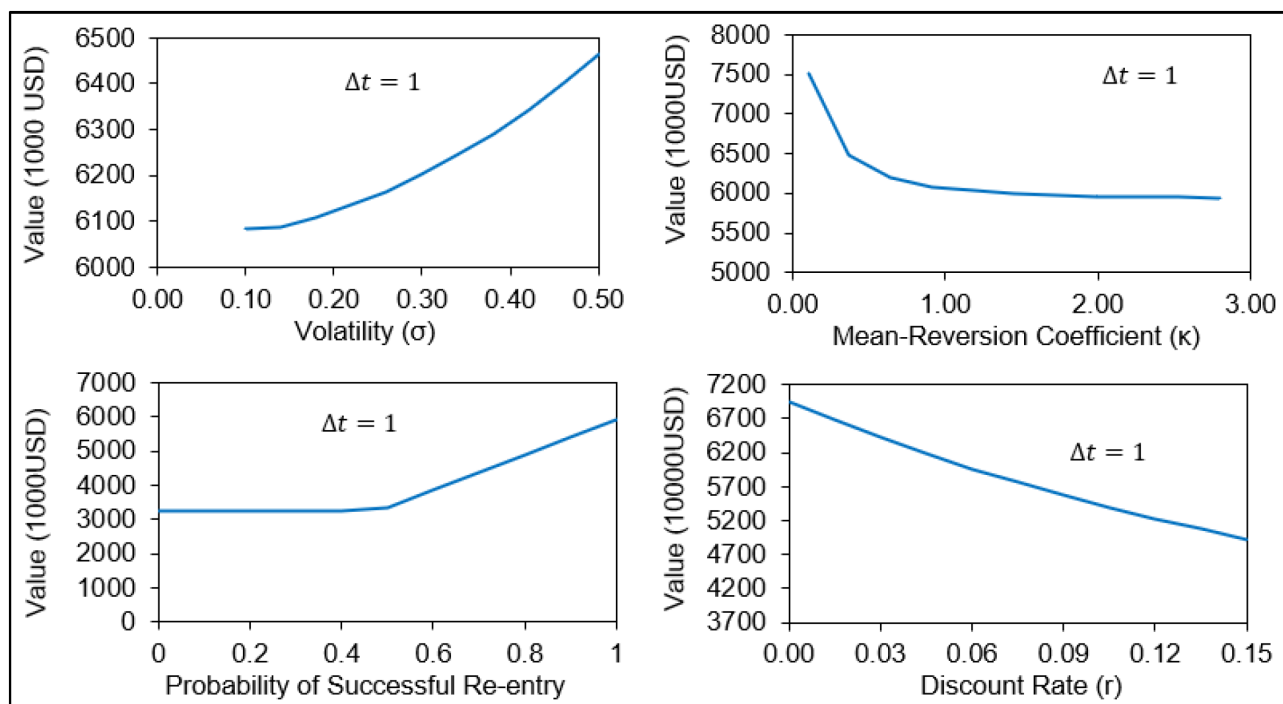


Figure 9—Two-dimensional sensitivity analysis that demonstrates the impact of volatility, mean-reversion coefficient, likelihood of success, and discount rate on total project value.

The three-dimensional sensitivity analysis is illustrated in Fig. 10. The figure shows the impact of the short (χ_0) and long-term (ξ) factors of the oil price model on the project value in a three-dimensional

sensitivity analysis. The short-term range is assumed to be -0.3 to $+0.3$, while the long-term range is considered to be 3.5 to 4.5 . The sensitivity analysis indicates that the oil price has a more considerable impact on the option value. Additionally, short-term fluctuations are especially influential at higher price levels. However, in a mean-reverting process, these fluctuations tend to decrease over time. As a result, for projects with a long production lifespan and delayed re-entry drilling, the effect of the short-term factor on the option value is likely to be weaker.

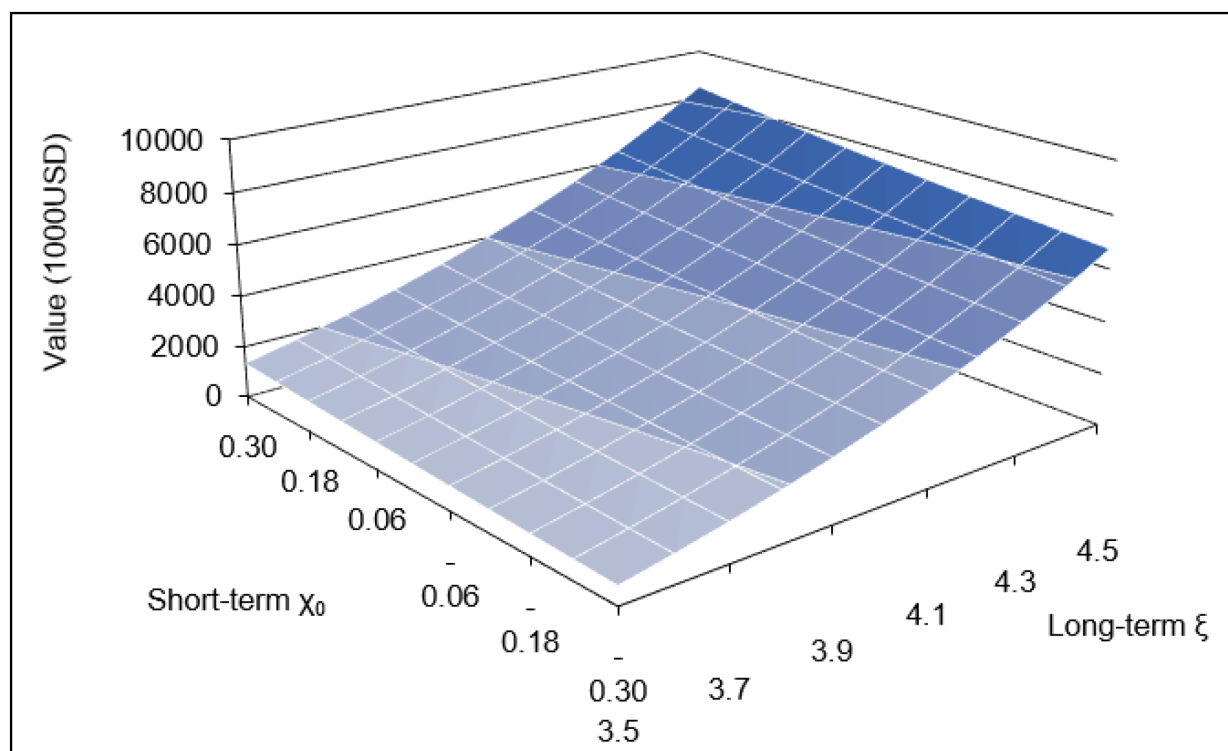


Figure 10—The three-dimensional sensitivity analysis of Schwartz-Smith's two-factor model, the mean reversion process's short—and long-term variables, with the total value.

The Conclusion

The decision to redevelop, abandon, or take no action on an unconventional mature field with steeply declining cash flow is challenging due to uncertainties of different natures, including geological and market-related factors. The unique characteristics of CTD make it an ideal tool for accelerating cash flow, specifically during prices upturn. The main value drivers of CTD are the possibility of accessing depleted reservoirs, improving recovery, and delivering a well in significantly less time and at a lower cost than conventional drilling rigs. However, failure to reach the target due to wellbore collapse is an inherent risk associated with this technology. Additionally, declining mature fields have a limited timeframe for CTD application, after which it may become economically unprofitable or technically unfeasible. Permanent abandonment is also an option when the future value of the field is unappealing. Making the right decision at the right time is essential for maximising the value.

This study offers a valuation model that considers oil price uncertainty and technical uncertainty expressed as the likelihood of reaching the required depth. It employs a fully risk-neutral approach with a real option framework. Additionally, it models oil price dynamics using a binomial lattice method that incorporates a mean-reverting process. A forward curve is generated at each node within the binomial lattice, enabling the estimation of the value of each potential decision. The dynamic decision-making process is modelled as a Markov chain, determining whether to proceed with declining value, re-enter drilling, or

opt for abandonment. Subsequently, a recursive algorithm is used to calculate the value of each option employing a risk-neutral approach.

The example shows that CTD can access the remaining reserves before the field hits its economic limit. This can maximise value, especially when benefiting from favourable oil prices. Since prices tend to revert to the mean, starting from a period of high oil prices (above the long-term average) makes re-entry more appealing, supporting the drilling option. Consequently, the field's value with an infill option is higher than without one. However, delaying the decision may be more advantageous when the current spot oil price falls below the long-term average. The discounted value of the abandonment cost affects the choice to postpone abandonment. In this case, the impact of the abandonment cost is minimal because it considers only one well, which reduces its influence. Other important factors in the decision include the probability of success in reaching the target, declining production rates of the original wells, and recoverable reserves. The proposed model simplifies these uncertainties, supporting more informed decision-making. This valuation model is notable for its simplicity and versatility in processing technical and economic inputs, delivering results within seconds. Additionally, the model features a sensitivity analysis of the key variables.

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